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Authors' address: Dr. HARTMUT GREVEN and Dr. GÜNTER CLEMEN, Zoologisches Institut der Universität, Hüfferstr. 1, D-4400 Münster, FRG

*Department of Biological Sciences, University of Montreal, and
College of Medical Technology, University of Tsukuba, Japan*

Qualitative and quantitative brain morphology in the Sirenian *Dugong dugong* Erxl.

By P. PIRLOT and T. KAMIYA

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Abstract

The brain of the Sirenian *Dugong* is described. In gross and qualitative morphology, it appears to be lissencephalic, with reduced olfactory areas and small olfactory and optic nerves but large sensory trigeminal components. In detail and quantitatively, its total size and its neocortex betray a much lower evolutionary level than those of the river dolphin *Platanista* taken for comparison. Its rhinencephalon is less reduced but its hippocampus is about as strikingly small as the dolphin's. Neither cerebellum nor mesencephalon point to any kind of specialization for aquatic life similar to that found in the dolphin. Although both animals are adapted to water by their general body morphology, the Sirenian seems to have remained far behind the Cetacean in its evolution in that direction from the viewpoints of progression and specialization.

Introduction

There exist very few accounts of the morphology of the Sirenian brain of which DEXLER's (1913) is probably the only detailed one. It deals with precisely the same species that we have here at our disposal (*Halicore dugong* = *Dugong dugong*). MIKLOUHO-MACLAY (1885) published a few remarks only on another species (*H. australis*) of the same genus.

DEXLER (1913) gave a number of particular measurements (thickness of nerves or cell layers; width of particular superficial areas, etc.) in the traditional fashion of his time but his lengthy and in general valuable descriptions do not entail any biometrical treatment in the modern sense. In his final comparative remarks, he simply takes recourse to the common and rather vague expressions "well developed", "medium size", etc. In his conclusion, he refers to the brain-to-body ratio in the dugong, in comparison with a very few other mammals, but no accurate figures are given and the author did make no use of other data published on this topic in his time by several European anatomists.

It therefore appeared to us that, while we do not need to add much to DEXLER's *qualitative* descriptions, we could significantly increase our *quantitative* knowledge of the Sirenian brain. This in turn seems to be timely since the usefulness of biometrical data on mammalian orders other than the commonly treated groups (Rodents and Primates, especially) has become increasingly obvious to morphologists over the past 25 years: the interest in wild, exotic species originates from a more enlightened concern for functional significance in the study of anatomical structures.

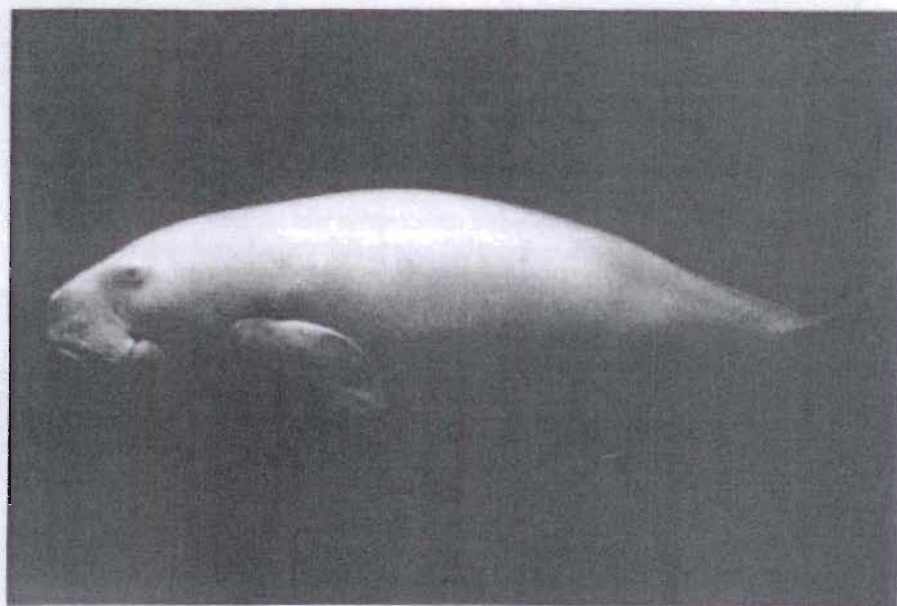


Fig. 1. *D. dugong*. Live individual in the Toba Aquarium, Japan

Materials and methods

Our material consisted of one brain of *Dugong dugong* from a female individual donated to the Japanese Government by the Indonesian Government in 1975, following the International Ocean Expedition which took place at that time in the Okinawa area. The animal's body length was 204 cm; its body weight was 155 kg and its brain weight 250 g. The brain was fixed *in toto* in 10% formalin. It was divided sagittally and one half was embedded in paraffin, sectioned at 15 μ m and Nissl stained. Photographs (116 in all) of sections at regular intervals from front to back were prepared and used for the delimitation of neuronal fields. The technique for estimating the volumes of each brain component has been explained in several publications (e. g. in STEPHAN et al. 1970).

The method of comparison is the same as that applied in previous papers by the present authors. It uses the progression indices the principle of which rests on a suggestion by WIRZ (1950) who proposed to refer the size of any mammalian brain to that of lowly, primitive Insectivores taken as basal types. The procedure can be summarized as follows (from KAMIYA and PIRLOT 1980, p. 108). "In brief, comparisons are made between the volumes of the brain (or brain components) in an idealized primitive type ('basal Insectivores') that would have the same body weight as the animal being studied. So, the allometrical relationship between brain and body is taken into account, which allows to describe and compare evolutionary stages in animals with various overall sizes. The basal value is 100 so that any figure above 100 indicates evolutionary progression while any figure below 100 betrays evolutionary regression".

Here are the abbreviations used throughout this paper for the brain components. N = neocortex; RH = rhinencephalon i. e. paleocortex with amygdaloid area; S = septum; D = diencephalon; St = striatum; H = hippocampus; M = mesencephalon; C = cerebellum; O = medulla oblongata.

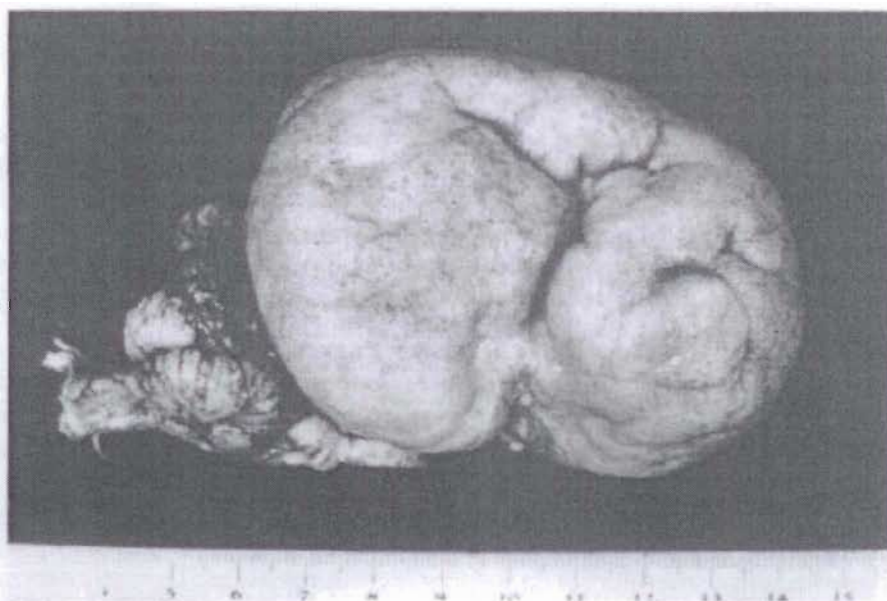


Fig. 2. *D. dugong*. Lateral view of brain. Description in text

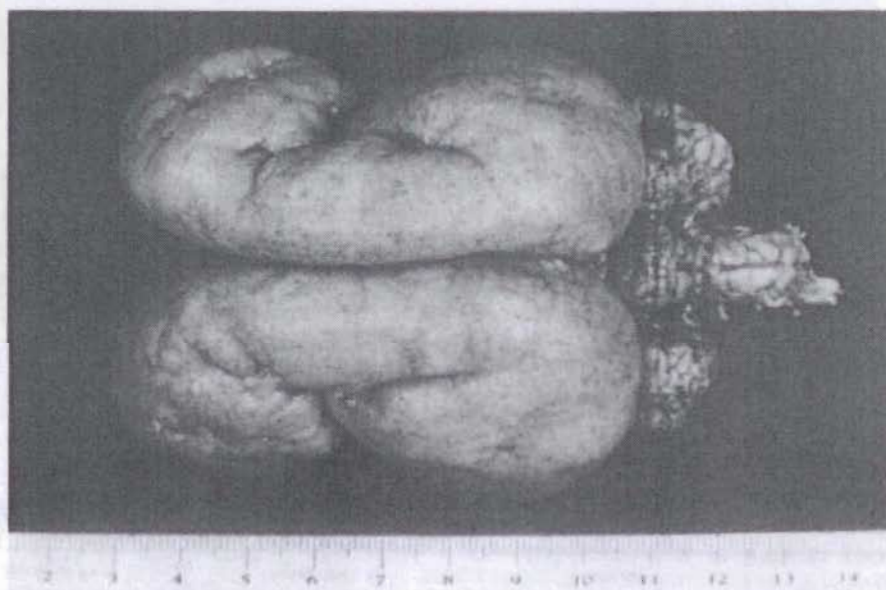


Fig. 3. *D. dugong*. Dorsal view of brain. Description in text

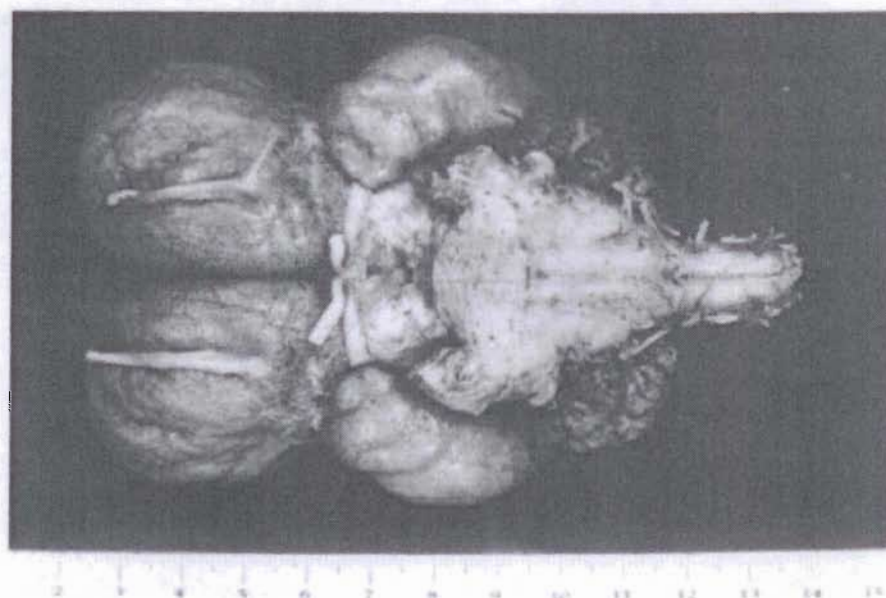


Fig. 4. *D. dugong*. Ventral view of brain. Description in text

Results

Qualitative surface morphology

We refer the reader to DEXLER (1913) for histological details. Only superficial features will be stressed here (Figs. 2-4). In *norma verticalis*, the cerebrum is remarkably square in outline, its two hemispheres appearing curiously kidney-shaped, as already noted by PETIT (1955). There is a deep Sylvian fissure giving off two longitudinal branches visible dorsally, one anterior and one posterior. The surface is smooth. The hemispheres slightly overhang the cerebellum. The floccular portions of the latter are well developed transversally. In *norma lateralis*, a short but well marked pre-sylvian fissure is visible. There is only a shallow depression on the latero-inferior plane homologous to a rhinal fissure. In *norma basalis*, the most striking feature is the small size and ribbon-like aspect of the two olfactory tracts that lie in well marked olfactory sulci but do not terminate in any bulbs. The optic nerves also are small (in comparison with those of the horse or pig, for example; see SISSON and GROSSMAN 1953). In contrast, the trigeminals are enormous in diameter.

Quantitative encephalization and brain composition

We shall consider successively the encephalization, the percentile subdivision of the telencephalon and the progression indices for each brain component. 1. Applying the method outlined above, we obtain a progression index of 313 for total brain in *Dugong*, a markedly lower value than the 842 found in our previous work for another aquatic mammal, the dolphin *Platanista* (KAMIYA and PIRLOT 1980). 2. Table 1 presents the quantitative composition of the brain of *Dugong* as well as the percentile composition of its telencephalon. 3. Table 2 provides the progression indices for the brain and the brain components in both *Dugong* and *Platanista*. 4. Table 3 gives a few examples of other encephalization indices for Ungulates and Cetaceans i. e. for herbivorous and for aquatic mammals, in order to help the reader appreciate the relative significance of the figures for *Dugong* and *Platanista*.

Table 1
Percentage composition of the *Dugong* brain

	Volumes	% total brain	% telencephalon
N	127460.42	57.17	78.16
RH	22549.38	10.11	13.82
S	1629.19	0.73	1.00
D	15643.63	7.01	—
St	9386.18	4.21	5.75
H	2074.54	0.93	1.27
M	2863.77	1.28	—
C	32651.51	14.64	—
O	8743.52	3.92	—
	223002.14	100.00	100.00

For abbreviations, see text.

Comments

On qualitative features

The first comparison that comes to mind is with some familiar, terrestrial Ungulates. In this respect, the most striking originality of the dugong's brain lies in its being lissencephalic, whereas a horse's or a cow's brain is gyrencephalic. The contrast is, of course, even greater

between dugong and dolphin. It is commonly admitted that lissencephaly is a primitive feature. As in both ungulates and cetaceans, the Sylvian fissure in the dugong is well marked.

Quoting the description of the brain of *Tursiops truncatus* by MORGANE et al. (1980) and referring to their beautiful illustrations, one may also note other similarities between the dolphin and the dugong: "pronounced antero-posterior shortening of the forebrain" – "blunting of the occipital lobe". This resemblance in the general shape may be characteristic for those two groups of aquatic mammals endowed with bulky and rounded heads; but to what extent it should be linked with living in the water, is unclear. Differences are also conspicuous: for example, nothing like the "large olfactory lobes" of *Tursiops* is seen in *Dugong* whose reduced olfactory tracts end in rather flat olfactory areas. The diagonal band area, however, seems to be well developed in both the dugong and the dolphin (MORGANE et al. 1980, Fig. 42).

Another characteristic worth emphasizing is the sensory equipment of the dugong, as seen from the cranial nerves (Fig. 1), in contrast with terrestrial ungulates. The olfactory bulbs are absent; the olfactory nerves are small and the optic nerves are also very modest, in comparison with what we observe in the horse, the cow and the pig all of which have large bulbs and large eyes. The trigeminal, however, is rather like that in the ungulates, i. e. fairly voluminous. Comparing now the dugong with the dolphin (*Tursiops*: MORGANE et al. 1980, Fig. 25), one sees that the olfactory nerve is absent in the dolphin although this animal possesses a well-marked olfactory lobe, as indicated above, whereas the dugong has a definite olfactory nerve (albeit a small one), but only a shadow of an olfactory lobe. This looks like a somewhat paradoxical situation. It is all the more intriguing since, as can be seen again from MORGANE et al. (1980), Fig. 29, another Cetacean, *Balaenoptera physalus*, has an olfactory lobe comparable in size with that of *Tursiops* (MORGANE et al. 1980, Fig. 25) but an olfactory tract very similar to that of *Dugong*. It thus seems that a diversity of arrangements and relative developments of those parts can be found in brains of aquatic mammals.

On quantitative results

The difference in encephalization between *Dugong* and *Platanista* (Table 2) is quite remarkable, as demonstrated by the progression indices. The ratio of the two figures $842/313 = 2.69$ seems to imply a considerable evolutionary advantage for the dolphin. In Table 3 we give examples of mammalian indices for a few ungulates and cetaceans. Herbivorous mammals are hardly expected to compete with cetaceans in brain development. However, *Platanista* is almost certainly one of the lowest ranking genera in this order, from that viewpoint. The obvious lag in encephalization exhibited by *Dugong* in comparison with all

Table 2
Progression indices in the dugong and the dolphin

	<i>Dugong</i>	<i>Platanista</i>
N	1121	3903
RH	208	123
S	182	132
D	294	719
St	319	739
H	51	99
M	100	432
C	246	671
O	133	325
Brain	313	842

Figures for *Platanista* taken from KAMIYA and PIRLOT 1980.

dolphins is likely to reflect a less progressive way of life, broadly conceived in terms of adaptiveness, mobility, defensiveness, sociability, etc. as is also suggested by the comparison of cortical development that follows. On the other hand, if one looks at the figures in Table 3, one can see the effect of negative allometry in that some gigantic species show very low indices (180 for a *Balaenoptera* weighing over 70 tons) although a less impressive though still bulky giant such as *Elephas* (3 tons) shows a much higher index (1044) than many smaller types. In fact, those figures as well as many others previously published (PIRLOT and STEPHAN 1970; STEPHAN et al. 1974; PIRLOT 1980) prove that the progression index does not vary uniformly with gross size but really expresses progress. It does so by referring to a standard para-ancestral form (see 'Materials and methods') which serves well the purpose of an evolutionary description, as was implied by WIRZ (1950) who followed this methodological principle. The low value 221 for *Tapirus* (weight 160 kg comparable to 155 kg for *Dugong*) is clear demonstration of difference in evolutionary level.

The percentile composition of the telencephalon in *Dugong* presents nothing unusual except for the surprisingly low value for the H contribution.

The progression indices for brain components show some interesting figures. The ratio of the neocortical indices for *Dugong* and *Platanista* is unexpectedly high: $3903/1121 = 3.48$, which is more than the corresponding ratio between encephalization indices, as seen above (2.69). From general knowledge, most biologist would probably say that a dolphin is "bound" to be more brainy than a sea-cow: there are so many nice stories about the dolphin's "mind-in-the-water"... Feelings of this sort can be misleading (PIRLOT 1980, on the lazy tree-sloth) but are probably correct in this case.

The RH indices differ considerably (208 for the dugong and 123 for the dolphin). The figure for the dolphin might have been lower if the animal used had been an adult. Both mammals are adapted to aquatic life, which is believed to result in strong decreases of olfactory structures and functions, but the Cetacea achieved that transformation millions of years before the Sirenians did; the latter may not have completed their evolution in that respect - if it means anything to see that as a "goal". Indeed, as late as the mid-Miocene, some Sirenians retained very developed olfactory components, including fair-sized bulbs (KAMIYA et al., in press). The important fact is that the living dugong, like the dolphin, has lost its subperipheral olfactory organ, the bulb, while retaining some remnants of olfactory tracts, contrary to some, if not all, dolphins. Size reduction of the more central components, RH in this case, did not proceed so far in the dugong as in the dolphin. Does this mean that the former has retained some vestigial olfaction whereas the latter has not? We cannot conclude at this stage. The RH includes components other than tissues involved in olfaction. Therefore, RH as we delimited it is a heterogeneous brain part, and our analysis fails to distinguish the fate of each element in it.

No comment will be made on S because of its anatomical uncertainty. It is obviously a poorly developed component in both animals.

The St is more progressive in the dolphin than in the dugong. This probably reflects its good correlation to the N, as shown by PIRLOT and JOLICOEUR (1982).

The H is amazingly reduced in the dugong and unprogressive in the dolphin, with progression indices of 51 in the former and 99 in the latter. Again, according to examples taken from the Primates (KAMIYA and PIRLOT 1980), it seems that H is generally smaller in more highly encephalized and neocorticalized species. We have also mentioned in the same paper that KRUGER (1966) found a very small H in several cetaceans. The possible relationship between hippocampal reduction and aquatic life is an open question.

With reference to basal Insectivores, the C stands much higher in the dolphin than in the dugong (671 against 246). Such an evolutionary increase in volume relative to a hypothetical ancestor may correspond to a superior control of locomotion in the former than in the latter. This interpretation is consistent with common observations of both animals in zoo pools, if not in the wild. The dolphin seems to be a much more agile swimmer than the dugong. One

may assume that elaborate swimming movements are related to a more developed C although it does not follow that acrobaticism by itself requires the largest C of all (PIRLOT and POTTIER 1977).

The M presents interesting data. First, if we compare the sets of three subtelenchalic components (D, M, O) in the two species, we find the following percentages for *Dugong*: D = 57.41; M = 10.51 and O = 32.08, whereas for *Platanista* D = 50.40, M = 18.16 and O = 31.44. Obviously, the difference between the two sets is that some relative volume pertaining to D in the dugong has so to speak shifted to M in the dolphin. The progression indices give a clearer view of the respective size increases with reference to the basal type. Curiously enough, the ratios of the indices for D and O in both animals are equal ($719/294 = 325/133 = 2.44$). But the ratio for M is $432/100 = 4.32$, which means that mesencephalic progression has been considerably greater during the Cetacean than the Sirenian history. One may suspect that this special enlargement of M in Cetacean evolution corresponds to the increase (relative to body-size) of the mesencephalic tissues devoted to acoustic high-frequency perceptions (sonar), for which the dolphins are famous. This hypothesis is strongly supported by previous studies on bats (STEPHAN and PIRLOT 1970; PIRLOT and POTTIER 1977). The very high index for D in the dolphin confirms the finding by PIRLOT and JOLICOEUR (1982) in bats that the progression of D is positively and strongly correlated with N. Apparently this is also the case in dolphins.

Conclusions

The brain of *Dugong*, which is the subject of the present study, appears to exhibit an *average degree* of progression when it is compared with a broad sample of brains of other mammals, that of man excluded because of its special development. In fact, we can try to refer the dugong to a pair of mammalian types, the herbivorous and the aquatic – it being an aquatic herbivore, an unusual status. A Sirenian can be expected, perhaps, to possess something fundamental in common with those two groups, i. e. something important in its general life-habits. So, the question is whether the brain of the dugong also shows any important common feature with those of the Cetaceans and Ungulates, especially in its quantitative aspects that are good indicators of evolutionary level. In its qualitative (gross) morphology, the Sirenian brain looks rather primitive because it is not gyrencephalic like that of the Cetaceans or most Ungulates. In this relative size, however, it is not so primitive as some

Table 3
Encephalization in Sirenians (A), Cetaceans (B) and Ungulates (C)
(Progression indices)

A. <i>Dugong dugong</i>	313	C. <i>Diceros bicornis</i> ²	245
B. <i>Platanista gangetica</i>	842	<i>Odocoileus</i> sp. ²	410
Four dolphins	675 to 2308	<i>Mazama</i> sp. ²	765
<i>Balaenoptera sibbaldi</i> ¹	180	<i>Antilocapra americana</i> ²	670
C. <i>Hippopotamus amphibius</i> ¹	158	<i>Tragulus javanicus</i> ¹	304
<i>Phacochoerus</i> sp. ¹	276	<i>Capra hircus</i> ¹	518
<i>Potamochoerus</i> sp. ²	273	<i>Oryx beisa</i> ¹	443
<i>Tapirus indicus</i> ¹	215	<i>Cephalophus maxwelli</i> ¹	532
<i>Tapirus terrestris</i> ²	221	<i>Camelopardalis giraffa</i> ¹	393
<i>Equus quagga</i> ²	505	<i>Okapia johnstoni</i> ²	477
<i>Equus przewalskii</i> ²	443	<i>Lama</i> sp. ²	500
		<i>Elephas indicus</i> ¹	1044

¹ From DUBOIS (1897). ² From KRUSKA (1973). Data for dolphins borrowed from KAMIYA and PIRLOT (1980).

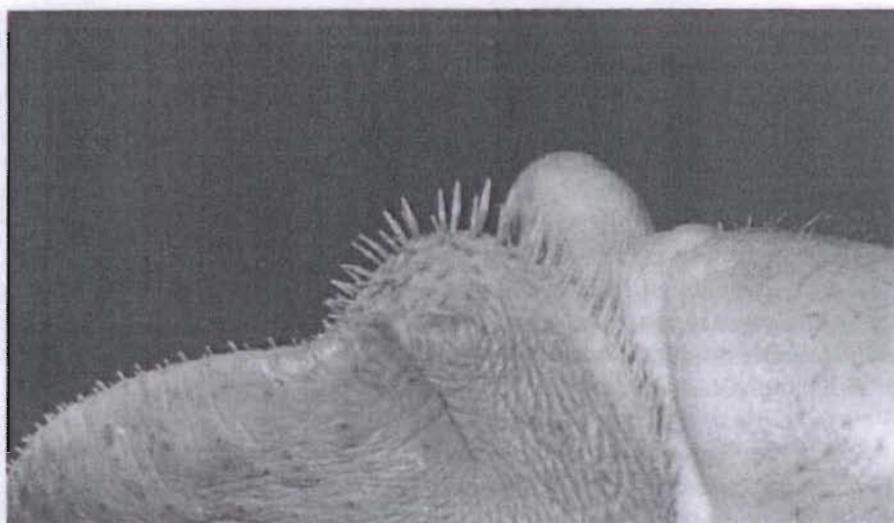


Fig. 5. *D. dugong*. Sinus hairs on snout. Anterior end of head to the left

Ungulate types are, as suggested by Table 3. One important observation that has already been commented upon by KAMIYA et al. (in press) is the presence of a very large trigeminal, with branches 1 and 2 especially developed: this feature no doubt betrays intensive sensory perceptions on the rostral part of the head and points to a very active foraging in the shallow coastal waters (Fig. 5, sensory hair).

Comparing the volumetric compositions of the brains of *Dugong* and *Platanista*, we see that important differences exist between them as far as the evolutionary development of the neocortex is concerned. Indeed, the progression index for the neocortex shows that the dugong is still far behind the dolphin in the evolution of its higher areas. In contrast, one may say that it is fairly close to the dolphin in the specialization implied in the strong reduction of its olfactory structures (bulb and rhinencephalon). On the other hand, the mesencephalic underdevelopment in the dugong probably points to the absence of extreme sensory specialization as suggested by the relatively large colliculi posteriores of the dolphin which is endowed with a sonar. Finally, the cerebellum of the dugong does not indicate a high degree of agility in the water, if compared with the dolphin's. Those observations justify a view of the dugong's brain as *moderately evolved* in a general way since, on the one hand, its neocortex exceeds in relative progression that of many mammals (Rodentia, Dermoptera: PIRLOT and KAMIYA 1982; Edentates: PIRLOT 1980; Fossil Sirenians: KAMIYA et al., in press; Chiroptera: PIRLOT and POTTIER 1977); on the other hand, it suggests neither a rich pattern for associations as required for intensive social life (as in many dolphins) nor a sophistication of the behavior in aquatic environment (sonar, acrobatic locomotion) as in the dolphin again. In fact, its most striking similarity with the Cetacean brain consists of the negative feature represented by the loss of the bulb and the extreme reduction of the hippocampus. Not a single trait in that brain points to an *advanced general progression* or to an *advanced particular specialization*. No promising adaptation or behaviour should be expected from a mermaid, whatever the legends about that kind of sea beauty may inspire in poets. What is needed, however, in order to substantiate further the above interpretation of the Sirenian brain, is more observations in the wild.

Acknowledgements

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Zusammenfassung

Qualitative und quantitative Gehirnmorphologie der Sirene *Dugong dugong* Erxl.

Die grobe Morphologie sowie quantitative Besonderheiten des Gehirns der Sirene *Dugong dugong* Erxl. werden beschrieben. Das Gehirn ist lissencephal, die olfaktorischen Areale sind reduziert, N. opticus und olfactorius sind klein, die sensorischen Trigeminusanteile groß. Sowohl in Einzelzügen der Gestalt als auch quantitativ, in der Gesamtgröße und jener der Neocortex zeigt sich, daß das Evolutionsniveau niedriger ist als beim Flußdelphin *Platanista*. Das Rhinencephalon ist beim Dugong weniger reduziert, aber der Hippocampus ebenso klein wie bei diesem Delphin. Weder Cerebellum noch Mesencephalon weisen irgendwelche Anzeichen einer Anpassung an das Leben im Wasser auf, die jenen des Delphins ähneln. Obwohl beide Tierarten in der allgemeinen Körperform Anpassungen an den Wasserhaushalt haben, bleiben die Sirenen in der Progression und Spezialisierung des Gehirns hinter den Cetaceen zurück.

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Authors' address: PAUL PIRLOT, Department of Biological Sciences, University of Montreal, Montreal H3C 3J7, Canada; TOSHIRO KAMIYA, College of Medical Technology, University of Tsukuba, Ibaraki 305, Japan